

Not All AI Agents Are the Same — So Why Do We Treat Them Like It?

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Originally Published: May 11, 2026 — LinkedIn

The Problem

We're no longer dealing with one kind of agent — we're dealing with an entire ecosystem. We have task-automation agents that handle workflows, healthcare agents that triage symptoms or coordinate care, and embodied agents inside humanoids, drones, and mobile robots. Calling all of them simply "agents" is like calling every animal in your home a "pet." Technically true, but not remotely specific.

That's why understanding what we mean by "agent" is no longer optional. The differences between these systems shape expectations, risks, governance, and how we deploy them in the real world.

The Argument

If we want to build and scale agents responsibly, we need a clearer language — and that's where the Agent Capability Maturity Model (aCMM) comes in.

We're all using the word "AI agent" — but we often mean completely different things. A task-automation agent that files tickets, an LLM tool-orchestrator, a multi-agent coordination system, and an autonomous mobile robot can all be labeled "agents" — yet their capabilities, risks, and governance needs are nowhere near comparable.

Existing Approaches

Several attempts have been made to define "agent maturity" ladders, including:

- Salesforce's Agentic Maturity Model — copilots → orchestration → multi-agent
- Agent Maturity Model by Samet Kilic, Technical Architect @ IBM — Helper → Integrator → Planner → Boss → Team
- 5 Levels of Agentic AI Intelligence by Shubha Pant and Mahesh Viswanathan @ Cisco — autonomy ladder
- Agentic AI Maturity by James M. Sims, Cognition Consulting Group — ad hoc → self-evolving systems
- AI Agent Taxonomy by Daniel W. Rasmus @ Serious Insight — argues agents can't be reduced to a single ladder

These frameworks are genuinely helpful — they give us shared language and a roadmap.

Why Linear Models Break Down

A single linear "level" breaks down almost immediately in real deployments. Teams building agents often see capabilities accelerate unevenly. An agent may advance quickly in areas like tool use, memory, or planning, while other dimensions — especially autonomy or learning — are deliberately constrained for safety, compliance, or operational reasons.

This asymmetry becomes even more pronounced with embodied agents such as robots and drones, or agents deployed in regulated environments. And as LLM-based agents gain self-improving behaviors, their capabilities no longer fit neatly into any single linear maturity ladder.

The Proposal: Agent Capability Maturity Model (aCMM)

A multi-dimensional capability-profile framework that evaluates agents across eight distinct dimensions instead of collapsing everything into a single maturity score:

1. Autonomy
2. Reasoning & Planning
3. Learning & Adaptation
4. Memory & Context
5. Tool Use & Integration
6. Collaboration & Social Ability
7. Embodiment
8. Domain Alignment

Each dimension is scored from 0–5, producing a capability fingerprint — a visual shape that makes trade-offs explicit and gives engineering, product, security, and governance teams a shared, evidence-based way to discuss what an agent can (and cannot) do.

Why This Matters

When you compare software agents in digital workspaces, embodied agents operating in physical space, medical and clinical AI agents, and mobile robotic or drone agents, you see completely different capability fingerprints — and suddenly the word "agent" stops being a single bucket and becomes a full spectrum of systems with very different risks, expectations, and governance needs.

Over time, this model can also expand to include domain-specific dimensions for sectors like healthcare, manufacturing, or finance, where specialized capabilities and constraints meaningfully shape how agents must be evaluated.

#AI #Agents #AgenticAI #aCMM #ArtificialIntelligence #Autonomy #AIEngineering #AIGovernance
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